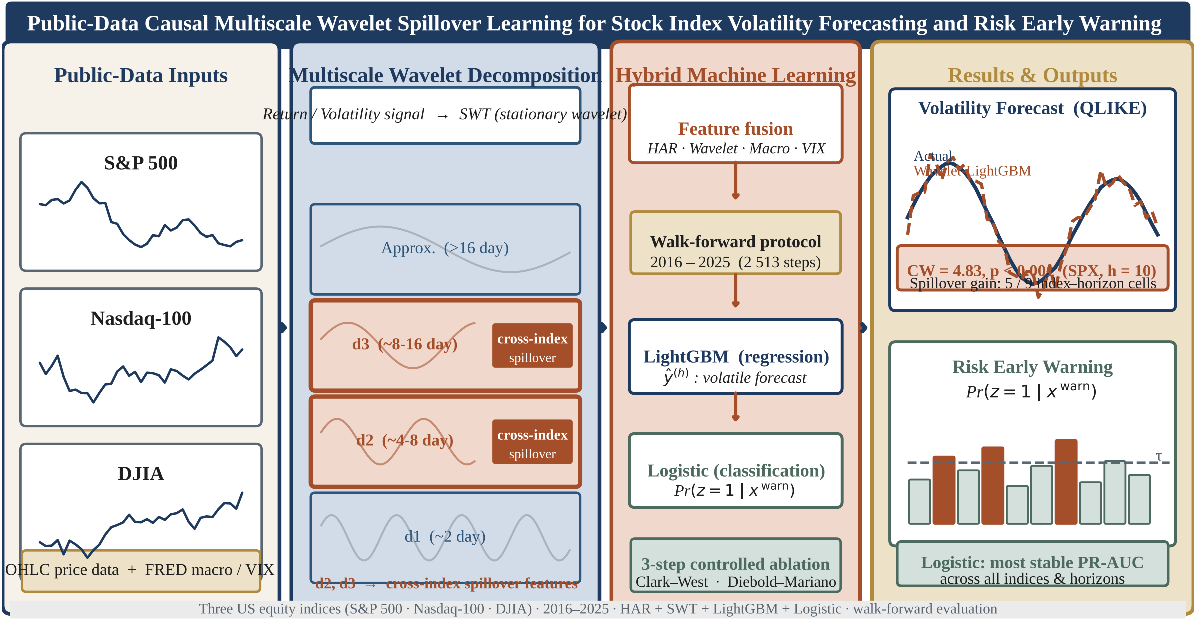

Public-Data Causal Multiscale Wavelet Spillover Learning for Stock Index Volatility Forecasting and Risk Early Warning
Public-Data Inputs
S&P 500
Nasdaq-100
DJIA
OHLC price data + FRED macro / VIX
Multiscale Wavelet Decomposition
Return / Volatility signal → SWT (stationary wavelet)
Approx. (>16 day)
d3 (~8-16 day)
cross-index spillover
d2 (~4-8 day)
cross-index spillover
d1 (~2 day)
d2, d3 → cross-index spillover features
Hybrid Machine Learning
Feature fusion
HAR · Wavelet · Macro · VIX
Walk-forward protocol
2016 – 2025 (2 513 steps)
LightGBM (regression)
$\hat{y}^{(h)}$: volatile forecast
Logistic (classification)
$Pr(z = 1 \mid x^{\text{warn}})$
3-step controlled ablation
Clark–West · Diebold–Mariano
Results & Outputs
Volatility Forecast (QLIKE)
Actual
Wavelet-LightGBM
CW = 4.83, p < 0.001 (SPX, h = 10)
Spillover gain: 5 / 9 index–horizon cells
Risk Early Warning
$Pr(z = 1 \mid x^{\text{warn}})$
τ
Logistic: most stable PR-AUC
across all indices & horizons
Three US equity indices (S&P 500 · Nasdaq-100 · DJIA) · 2016–2025 · HAR + SWT + LightGBM + Logistic · walk-forward evaluation